Akash Dubey

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Education

Rutgers University Honors College, School of Arts and Sciences — New Brunswick, NJ

Bachelor of Science in Computer Science and Mathematics — *Anticipated May 2027*

- **GPA**: 3.936/4.0 | SAT: 1580
- **Relevant Coursework**: Tensor Networks, Systems Programming, Stochastic Processes, Computer Architecture, Algorithms, Data Structures, Honors Calculus III & IV, Math Theory of Probability.
- **Academic Programs & Research**: Directed Research Reading Program on Symmetric Functions (representation theory for combinatorial optimization); Physics Learning Theory Seminar (statistical mechanics in high-dimensional learning), Dean's List.
- **Activities**: Quantitative Finance Club, Road to Silicon Valley Program (Cohort 6), ESL Instructor, Pickleball.

Engineering Experience

New York Life Insurance — New York, NY

Software Engineer Intern, TDAV Team — *May – Aug 2026*

- Created an AI text and voice agent for processing insurance claims for clients and 12,000 Customer Service Representatives using AWS Infrastructure with access to NYL services, knowledge, and forms.
- Constructed a company wiki/knowledge base from 13,000 documents compiled from Sharepoint, Public Site, Confluence, Jira, and the Intranet bringing business logic into one place for AI and human users.
- Worked on multi-agent systems hosted on NYL's Cloud responsible for improving website development in AEM, Pega code, Tungsten document processing, and mainframe modernization.

Scarlet Sync — New Brunswick, NJ

Founder — *Jan 2025 – Present*

- Built a modern Rutgers academic planning platform with thousands of users for degree, schedule, and class planning as well as campus discovery, utilizing AI-assisted parsing of unstructured data from legacy university services.
- Architected a robust data ingestion pipeline using Python and LLMs to reverse-engineer legacy data, indexing over 2,500 classes and 500+ degree programs for real-time querying.
- Developed a production AI assistant with streaming tool-calling and grounded retrieval over schedules, degree plans, and profile context, enabling automated schedule building and personalized academic planning across Web, ChatGPT, Claude, and iOS.

Lykke — San Francisco, CA

Founding Software Engineer — *Sep 2025 – Present*

- Built core AI workflows for an AI study platform used to ingest Canvas LMS courses, assignments, announcements, notes, and uploaded documents into class-scoped knowledge bases for contextual chat and study generation.
- Engineered adaptive RAG pipelines across MongoDB, Zilliz/Milvus, S3, Gemini Files API, and SSE streaming, including source inference, citation events, token-budgeting for large course materials, and fallback paths for extracted text vs. full-file reasoning.
- Created a distributed course-wiki generation engine that crawls educational sources, synthesizes structured sections, generates interactive artifacts, embeds wiki content for retrieval, and runs tenant-isolated background builds via SQS, MongoDB leases, Redis rate limiting, and AWS ECS.

Samaritan Scout — Cranford, NJ

Full Stack Engineer Intern — *May 2023 – Aug 2025*

- Developed an agentic scraping system leveraging OpenAI and Gemini structured outputs to autonomously extract data from over 100,000 non-profit websites, reducing manual data entry time by over 90%.
- Developed a full-stack search engine using React, TypeScript, and PostgreSQL to index 35,000+ volunteer opportunities, implementing semantic search algorithms with hallucination filters to ensure accurate matching.

Research

Rutgers Economics Labs — New Brunswick, NJ

President (formerly Research Director, Team Lead) — *Oct 2024 – Present*

- Leading quantitative research projects for partners including NJDOL, NJDEP, NJBPU, Federal HUD, U.S. Census Bureau; applied econometric models in Python/R to analyze large datasets and project economic trends.
- Architected an AI automated economic research pipeline RAIL (Rutgers Agentic Intelligence Labs) to collect, organize, analyze, and synthesis economic data to derive insights using data ontologies and coding agent workflows
- Previously led a team of five researchers to evaluate the efficacy of the Urban Enterprise Zone program for the NJDCA, utilizing econometric methods in R and Python to analyze complex census datasets.

Algorithmic Robotics and Control Lab in Rutgers's Dept. of CS — New Brunswick, NJ

Research Assistant (Advisor: Prof. Jingjin Yu) — *Jan 2026 – Present*

- Implemented and validated CUDA versions of pRRTC and FCIT*, parallelizing tree/graph search and collision checking for high-dimensional ASAO robot motion planning across 500 benchmark problems.
- Developing a graph-based search policy (GaP) for dual-arm task-and-motion planning (TAMP), learning from successful and failed planner traces to rank symbolic action skeletons and predict geometric feasibility. Working in Isaac Sim specifically in dual-arm motion planning to further bridge TAMP capabilities with ML.

Kwan Lab, Rutgers Cancer Institute of New Jersey — New Brunswick, NJ

Computational Biology Research Assistant (Advisor: Prof. Kelvin Kwan) — *Jan 2026 – Present*

- Developed an AlphaGenome-based in silico deletion screen spanning 12 candidate sites, 3 deletion sizes (10, 40, and 80 kb), and 2 cell types (72 perturbation conditions) to quantify TAD-boundary sensitivity; identified a chr12 insulator with the strongest predicted insulation loss among 12 candidates.
- Built a 3D chromatin polymer simulator using overdamped Langevin dynamics to generate conformations from Hi-C contact matrices; implemented multi-scale TAD detection using insulation score and directionality index methods and validated the pipeline with 70+ automated tests.
- Configured and validated a reproducible DeepSTARR inference environment, resolving legacy TensorFlow/Keras dependencies in HPC to enable sequence-based prediction of developmental and housekeeping enhancer activity.

Selected Technical Projects

Knowledge Base and AI Agent Workflow Manager | *Data Ontology, Workflows, Coding Agents*

- Built KRAIL (pypi), a local-first, repo-backed knowledge runtime that structures raw notes into ontology-aware topic pages, markdown graphs, source ledgers, claim records, verification runs, and artifact lineage for evidence-grounded agent memory.
- Implemented MCP/CLI and agent infrastructure for auditable coding-agent workflows (for software factories, auto-research, and company brain use cases), including local runner dispatch, dry-run work orders, session logs, role prompts/checklists, workflow validation, and hybrid keyword-graph-vector retrieval.

Local Coworker: Native macOS Computer-Use Agent | *Swift, MCP, Gemini Vision* (private code)

- Built a native desktop agent capable of autonomous computer use by integrating Gemini Vision with macOS Accessibility APIs, AppleScript, and JXA.
- Implemented the Model Context Protocol (MCP) to orchestrate specialized sub-agents (UI Interaction, Shell, App Research) for planning and executing complex multi-app workflows.
- Engineered a "Research Agent" that expands the system's capabilities by autonomously searching documentation to learn new automation patterns and add them to its skill library.

SLM Optimization & Post-Training (GSM8k) | *PyTorch, HPC*

- Fine-tuned small language models (SLMs) on the GSM8k math reasoning dataset to see the effects of different model sizes and architectures on math reasoning capabilities.
- Deployed models on the Rutgers High Performance Computing (Amarel) cluster, utilizing quantization and structural pruning to maximize inference throughput on constrained compute.

Awards & Skills

Awards: 8090 AI Top Coder (5th), Rutgers Shark Tank (2nd and 3rd), ASA Fall Data Challenge (Top 3), NSF I-Corps.
Languages: Python, C++, CUDA, Java, SQL, TypeScript, R, Bash, Git
Technologies: PyTorch, Agents, Transformers, Generative AI, Pandas, AWS, Scikit-learn, HPC Clusters